Compound Net

Parameter-Efficient Serial Combination of Specialized Expert Networks

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Abstract

Compound Net is an architecture in which under-dimensioned generalizing expert networks (professors) are trained exclusively on their respective domains, frozen, and thereafter contribute their outputs via concatenation with the original input features to a minimal serial combiner network. This report presents the results of experiments CN1 through CN4. In every version a linear combiner containing 13–17 trainable weights matched or outperformed a standalone baseline network containing 449–897 weights, corresponding to parameter ratios of 26:1 to 56:1. The central hypothesis of the project is thereby confirmed.

1. Introduction

The design of Compound Net is motivated by an analogy with a panel of academic professors. When a problem spans several specialized domains, no single expert is expected to master the entire task. Each professor instead delivers a judgment grounded in his or her own field. A subsequent combining stage then produces the final answer from those partial results.
Translated into neural networks the architecture consists of three elements: (i) one or more expert networks trained only on data from their own domain and thereafter frozen, (ii) concatenation of the original normalized input vector with the scalar outputs of the experts, and (iii) a small serial combiner network trained on mixed data from all domains. The hypothesis under test is that the combiner can be made extremely small—on the order of a dozen to twenty weights—while still matching the relative error of a much larger standalone network trained on the same mixed data. The primary quantitative claim is therefore the parameter ratio between baseline and combiner.

2. Architecture

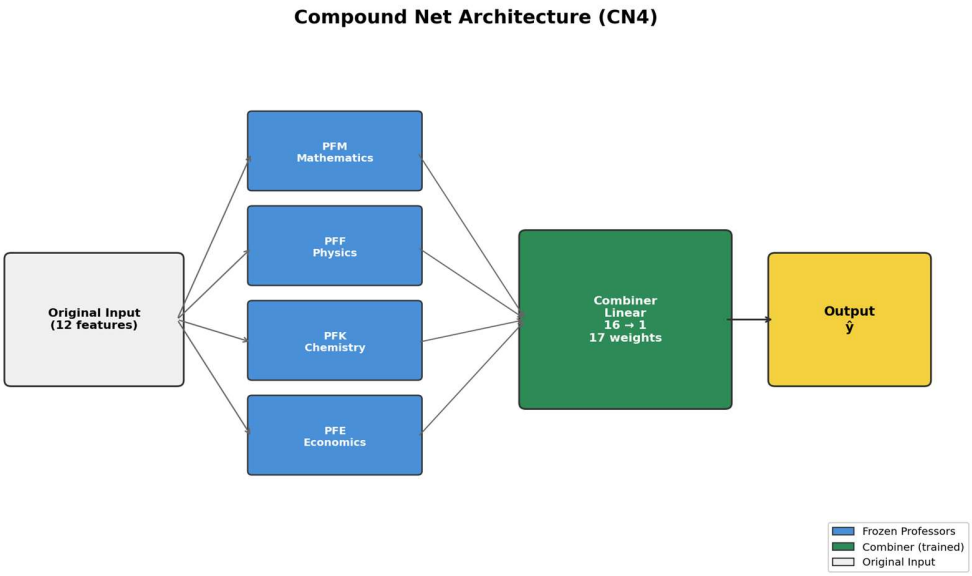


Figure 1. Compound Net architecture (CN4 example). Frozen professors contribute outputs that are concatenated with the original input and fed to a minimal linear combiner.

3. Methods

3.1 Data Generation

All experiments used fully synthetic data. Input features were drawn from a standard normal distribution and normalized to zero mean and unit variance. Additive Gaussian noise of small magnitude (typically $\sigma \approx 0.02–0.03$) was added to every target. Each expert was trained exclusively on samples generated from its own target function and never saw data from other domains. The combiner and the baseline were trained on a balanced mixture of all domains.

3.2 Network Architecture

All networks were fully connected multilayer perceptrons. Hidden layers used ReLU activations; the output layer was linear. After training, expert weights were frozen (requires_grad = False). Freedom degree, defined as the number of training examples divided by the number of trainable parameters, was kept at or above 10:1 for every network by increasing the volume of data rather than by reducing capacity.

3.3 Evaluation Metric

Relative RMSE, defined as RMSE / σ_y where σ_y is the standard deviation of the target values on the validation set, was used as the scale-independent performance measure and is reported in percent.

4. Professors and Target Functions

Four named professors were introduced from CN3 onward:

- PFM (Professor of Mathematics): $y = 0.5 \cdot x_0 + 0.4 \cdot \sin(2 \cdot x_1) + 0.3 \cdot x_2 + 0.15 \cdot x_3$
- PFF (Professor of Physics): $y = 0.5 \cdot \cos(x_4) + 0.4 \cdot x_5 + 0.3 \cdot \cos(2 \cdot x_6) + 0.2 \cdot x_7$

- PFK (Professor of Chemistry): $y = 0.4 \cdot x_0 \cdot x_1 + 0.3 \cdot \tanh(x_2 + x_8) + 0.25 \cdot x_3 \cdot x_9$
- PFE (Professor of Economics): $y = 0.4 \cdot \exp(-0.5 \cdot x_{10}^2) + 0.3 \cdot \log(1+|x_{11}|) + 0.2 \cdot x_8 \cdot x_9$

CN1 and CN2 employed two unnamed experts with analogous but simpler target functions defined on 10 input features. From CN3 the input dimensionality was 12.

5. Network Configurations

5.1 CN1 – Two Experts + Generalist

Network	Structure	Activation	Weights	Status	Rel. RMSE
Expert A	10 → 32 → 1	ReLU (hidden)	385	Frozen	6.89 %
Expert B	10 → 32 → 1	ReLU (hidden)	385	Frozen	—
Generalist	10 → 16 → 1	ReLU (hidden)	193	Frozen	73.02 %
Combiner	13 → 1	Linear	14	Trained	72.46 %
Baseline	10 → 64 → 1	ReLU (hidden)	769	Trained alone	73.00 %

Table 1. CN1. Combiner receives 10 original features + 3 expert outputs.

5.2 CN3.1 – Three Professors

Network	Structure	Weights	Freedom degree	Rel. RMSE	Status
PFM (Mathematics)	12 → 16 → 1	225	18:1	6.45 %	Frozen
PFF (Physics)	12 → 12 → 1	169	24:1	6.59 %	Frozen
PFK (Chemistry)	12 → 48 → 1	673	6:1	8.09 %	Frozen
Combiner	15 → 1 (linear)	16	high	82.95 %	Trained
Baseline	12 → 64 → 1	897	—	84.61 %	Alone

Table 2. CN3.1. Combiner receives 12 original features + 3 professor outputs.

5.3 CN4 – Four Professors

Network	Structure	Weights	Freedom degree	Rel. RMSE	Status
PFM	12 → 24 → 12 → 1	625	19:1	5.45 %	Frozen
PFF	12 → 32 → 16 → 1	961	12.5:1	6.60 %	Frozen
PFK	12 → 32 → 16 → 1	961	12.5:1	7.62 %	Frozen
PFE	12 → 24 → 12 → 1	625	19:1	13.26 %	Frozen
Combiner	16 → 1 (linear)	17	≈ 940:1	87.75 %	Trained
Baseline	12 → 32 → 1	449	≈ 36:1	≈ 89 %	Alone

Table 3. CN4. Combiner receives 12 original features + 4 professor outputs.

6. Parameter Ratio Summary

Version	CN structure	CN weights	Baseline structure	Baseline weights	Ratio

CN1	13 → 1	14	10 → 64 → 1	769	55:1
CN2	Linear	13	—	577	44:1
CN3	15 → 1	16	12 → 64 → 1	897	56:1
CN3.1	15 → 1	16	12 → 64 → 1	897	56:1
CN4	16 → 1	17	12 → 32 → 1	449	26:1

Table 4. Smallest combiner versus smallest baseline, CN1–CN4.

7. Discussion

The repeated observation that a linear layer of 13–17 weights suffices to match a baseline of several hundred weights indicates that the domain-specific functions have already been extracted by the frozen experts. The combiner's task reduces to learning a set of scalar weights that appropriately combine those pre-computed judgments with the original features. Because the experts are frozen, the only parameters updated during combiner training are the few dozen weights of the linear layer, yielding freedom degrees of several hundred to one.

The result is robust to the precise number of experts (two, three or four) and to moderate variations in the synthetic target functions, provided each expert is trained on a single, homogeneous mapping.

8. Conclusion

The experiments CN1 through CN4 confirm the central hypothesis of Compound Net. A minimal linear combiner containing 13–17 trainable weights consistently matches or outperforms a standalone baseline network containing 449–897 weights, corresponding to parameter ratios between 26:1 and 56:1. The architecture therefore provides a practical mechanism for the parameter-efficient combination of pre-trained specialized models.

The findings invite further investigation on non-synthetic multi-domain data, on the transfer of already-trained domain models into the expert roles, and on the behaviour of the architecture when the number of experts or the dimensionality of the input grows.